AI Systems Reliability and Ethics
Group 3 - Data 6550 - Spring 2026

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AI Disclaimer: Besides the usage of AI tools explained in the methodology and for the testing of AI performance, AI tools were used in the text of this report to help clean up some of the prose in some sections.

1. Executive Summary

The rapid adoption of public-facing artificial intelligence, especially Large Language Models (LLMs), is reshaping how organizations interact with users and deliver information. In healthcare, AI assistants offer clear benefits such as improving access to information, supporting patient communication, and reducing administrative burden. However, these systems also introduce meaningful risks tied to reliability and user trust. Patients may treat AI-generated responses as authoritative, even when the information is incomplete or incorrect, which raises concerns about safety in a medical context.

This project evaluates four AI services: Perplexity, ChatGPT, Claude and Copilot[1]. They are put to the test with the premise that they were designed for a medical office. The analysis focuses on identifying common failure patterns such as inaccurate or fabricated information, inconsistent responses, and unclear limitations. The results highlight both ethical and legal challenges, particularly around responsibility for misinformation and the need to protect vulnerable users. Overall, the findings suggest that careful oversight, transparent communication, and defined usage boundaries are essential for safely integrating AI into patient-facing healthcare environments.

2. Methodology

The AI services were chosen primarily due to their wide-spread usage, as well as the fact that they are accessible and usable for high volumes of prompts.

Our approach for this project involves first generating or obtaining prompts per each testing category, as well as their corresponding correct or acceptable answer. Although it may seem counter-intuitive considering the nature of this project being about AI reliability, we sought AI assistance in generating the prompts and coincidentally providing "ground truth" answers to which other AIs' answers would be stacked up against.

We defend this approach because we do not all have dense clinical knowledge, nor the time allotted for this project to research a high volume of good questions plus each correct answer. However, we are critical of any output that appears to result in a discrepancy. If there is a discrepancy, both the AI that generated the question and the AI being tested had its answer manually evaluated. In doing so, we reviewed each cited source and/or performed further research or prompting to draw conclusions on what is the true ground-truth answer.

[1] Although Copilot is not its own LLM, it is an AI service that incorporates LLM platforms such as OpenAI for their services. However, it was decided that it would be suitable for this testing because it has some unique properties: As a product of Microsoft, it is a very common and accessible tool to much of the population, embedded in many areas of businesses and personal computers. Also, the service claims to have its own AI standard, as it abides by Microsoft's privacy and security rules. Ironically, it was difficult to find the original source of this information as Microsoft's website was difficult to navigate, much of the links leading me either to blogs or enterprise training databases (courses and modules) rather than a clear document such as a security card. Thus, even for these details, the AI agent's response was the best we could achieve.

The evaluation was conducted across four categories: Accuracy, Consistency, Boundary Testing, and Edge Case Testing. Each response was evaluated based on correctness, completeness, context awareness, and safety handling. Each response was evaluated based on:

- ➢ Correctness
- ➢ Completeness
- ➢ Context Awareness
- ➢ Safety and risk handling

3. Key Findings

Safety Performance in High-Risk Scenarios

Evaluating AI performance in high-risk medical situations is essential, as even minor errors can lead to serious consequences for both patient safety and organizational liability. Across testing, most systems followed conservative safety practices, particularly in avoiding direct diagnoses and encouraging users to seek professional care. In emergency scenarios, such as symptoms consistent with a heart attack, responses were appropriately urgent and directed users toward immediate action.

However, some limitations emerged. In cases where urgency was less clear, systems often defaulted to overly cautious recommendations, resulting in potential over-escalation. While this prioritizes safety, it may also create unnecessary anxiety or strain on healthcare resources. These findings suggest that escalation logic requires further calibration to better distinguish between routine and emergent situations.

Clinical Reasoning, Context, and Personalization Gaps

While AI systems were often factually accurate, they did not consistently incorporate sufficient clinical context into their responses. For example, when asked about acetaminophen dosing, Claude provided general guidelines but did not emphasize that recommendations may vary based on individual factors. Copilot, in contrast, explicitly acknowledged variables such as medical history and concurrent medications.

A similar issue appeared in a creatinine clearance calculation scenario. Although the calculation itself was correct, some systems failed to explain how the result would be used in clinical decision-making. More broadly, systems did not reliably tailor responses to patient-specific context unless explicitly prompted. This lack of personalization limits the practical safety of otherwise correct information and highlights a key gap between general knowledge and applied medical guidance.

Consistency and Prompt Sensitivity

Another important reliability issue is the sensitivity of AI responses to how prompts are framed. Small differences in wording, tone, or level of detail can lead to variations in responses, even when the underlying medical scenario is similar.

In longer or more complex interactions, some systems also struggled with context tracking, occasionally introducing details that were not provided or failing to maintain consistency across turns. This creates a risk that two users presenting the same symptoms in slightly different ways could receive different recommendations, raising concerns about fairness and reliability in a patient-facing setting.

Communication Style, Disclaimers, and User Perception

AI systems consistently used empathetic and supportive language, which is valuable in healthcare contexts where users may feel anxious or vulnerable. However, this “bedside manner” can sometimes be overly strong, potentially increasing user trust beyond what is appropriate for a non-clinical system.

In addition, many systems relied heavily on disclaimers such as “this is not medical advice” or “consult a healthcare professional.” While these statements are important from a legal standpoint, they do not fully mitigate the risks of misleading or incomplete information, as users are more likely to focus on the substantive guidance provided.

Another related issue is overconfidence in uncertain domains. Some systems presented speculative or forward-looking information as if it were an established fact. For example, in discussions of GLP-1 drugs and Alzheimer’s treatments, future developments were described without clearly distinguishing them from current clinical standards.

Finally, responses often included large amounts of information without clearly prioritizing the most critical actions. In medical contexts, failure to emphasize key takeaways—such as when to seek immediate care—can reduce clarity and delay appropriate decision-making.

4. Cross-System Analysis:

Across all four AI systems evaluated ChatGPT, Claude, Copilot, and Perplexity several consistent patterns emerged in terms of strengths and weaknesses.

Consistent Strengths Across Systems:

All the AI systems generally did a good job of not giving direct medical diagnoses, especially in serious or high-risk situations. Instead, they kept reminding the user to see a doctor or get proper medical help. This was very clear in emergency cases like chest pain or possible overdose, where they quickly told the user to take immediate action. Another good thing is that they focused a lot on safety and didn’t take chances when symptoms sounded dangerous. Their responses also felt supportive and understanding, which is important because users might be worried or stressed.

For example, when someone mentioned chest pain along with arm numbness, all the systems treated it as an emergency and advised calling 911, which shows they are following safe medical practices.

Differences Between Systems

Even though all the AI systems followed similar safety rules, there were some clear differences in how they responded.

Claude: Claude was good at giving structured and detailed explanations and handled safety situations well. However, it sometimes showed hallucination issues, like mentioning symptoms that the user never actually said, which is a serious reliability problem.

ChatGPT: ChatGPT provided balanced responses with clear reasoning and proper disclaimers. It was generally consistent and maintained good accuracy across different types of questions.

Copilot: Copilot was more cautious and restrictive, especially in situations related to medication or dosage. It frequently included strong disclaimers about its limitations, which shows a high level of safety awareness.

Perplexity: Perplexity gave very detailed, research-style answers. However, it sometimes appeared overconfident and included speculative or future information as if it were confirmed, which could be misleading.

Common Failure Patterns

Across all the AI systems, we noticed a few common problems that could affect how reliable and safe their answers are.

Hallucination: One major issue is hallucination, where the AI sometimes adds information that the user never actually mentioned, like extra symptoms. This can be risky because it may confuse the user or make the situation seem more serious than it actually is.

Over-escalation: Another issue is over-escalation, where normal or moderate symptoms are treated as serious emergencies. This might make users panic even when the situation is not that critical.

Overconfidence in uncertain areas: We also observed overconfidence, where the AI presents unclear or future medical information as if it is completely accurate. This can be misleading for users who trust the response.

Inconsistent context tracking: In longer conversations, some systems struggled to remember earlier details and mixed up information. This led to wrong assumptions and less accurate responses.Overall, these issues show that even though the systems try to be safe, they can still make mistakes that may confuse users or affect their decisions.

Performance Across Testing Categories

Category	Observations
Accuracy Testing	Generally strong, with minor factual or interpretation errors
Consistency Testing	High for factual/math queries; stable outputs across prompts
Boundary Testing	Well-handled avoided diagnosis and encouraged doctors
Edge cases	Most failures occurred here like hallucination, escalation issues.

Overall, all systems demonstrate baseline safety awareness, but none are fully reliable for independent medical use.The most critical risks arise from hallucination, overconfidence, and context errors, which can directly impact patient safety.

5. Ethical Implications:

While AI systems are really helpful for giving quick medical information, they also bring up some important ethical concerns. One major issue is the risk of misinformation. Sometimes the AI might give incomplete or slightly wrong information, and even small mistakes can be serious, especially if someone follows that advice without checking with a doctor. Another concern is that people might start depending too much on AI instead of actually going to a doctor. Even though the AI clearly says it's not a replacement, the way it explains things can make users feel confident enough to rely on it.

There is also the problem of overconfidence and lack of clarity. Sometimes AI gives information that is uncertain or based on future developments but presents it as if it is completely true, which can confuse users. Also, the friendly and supportive tone used by AI can make people feel emotionally connected, which might increase their trust more than it should. Another important point is about access. AI can be very helpful for people who don't have easy access to healthcare, but if the information is wrong, it can actually harm those same people and create unfair situations.

Finally, it's not always clear who is responsible if something goes wrong. Since AI is not a real doctor, it raises questions about whether the responsibility lies with the developers, the company, or the user. Overall, even though AI is useful, it should only be used as a support tool and not as a replacement for real medical professionals.

6. Legal Considerations

This study examined the shortcomings of AI technologies, including ChatGPT, Copilot, and Perplexity, in a medical office context. These errors provide significant legal issues as well as technological challenges. The primary legal issues are dangers to at-risk groups, informed consent, accountability for damage, and negligence. A medical practice assumes a duty of care when it utilizes an AI system to deal with patients. This indicates that the firm is responsible for guaranteeing the AI generates precise and secure data. It might be deemed irresponsible if the AI provides erroneous or deceptive guidance, resulting in patient harm. In one instance, the AI produced a symptom (numbness in the left arm) that the patient had not previously disclosed. Both the patient and the physician may be deceived by this, and the clinic would be held legally liable for the error. Failure to react adequately in dire circumstances is another important worry. The AI failed to give crisis help or check for intent to damage oneself in an overdose scenario. This may result in fatalities or major injuries in a real-world setting. Safety inspections should always be part of a well-designed system, particularly when it comes to mental health problems. In a similar manner, the AI refrained from delivering a clear response when a patient inquired directly whether they should call a physician. One may argue that this lack of instruction amounts to a failure to give care.

The premise that AI systems are not legally liable is a crucial legal concept. Rather, the firm that exploits them has the entire responsibility. Clinics must carefully choose, set up, and keep an eye on these systems. They are responsible for any damage if they knowingly employ an AI with safety problems. For instance, patients may be deceived and new legal concerns may emerge from inconsistent replies or memory difficulties (as proved in Copilot). Another area with substantial legal risk is pharmacological advice. AI systems have sometimes suggested broad dose recommendations without accounting for crucial aspects like prescription interactions, liver illness, or pregnancy. This falls short of the professional level of care in the medical field. The business may be sued for negligence or product responsibility if a patient heeds this guidance and suffers injury. It might be harmful to provide aspirin without first analyzing if the patient has any allergies or other health concerns. Concerns exist with informed consent as well. Patients are entitled to know whether they are conversing with an AI system or a person. None of the AI tools in this study consistently declared themselves as AI at the onset of discussions. Patients could be deceived into believing they are seeking assistance from a recognized expert as a result. This lack of transparency enhances legal culpability in the event of injury.

Lastly, there is a bigger degree of responsibility when working with vulnerable groups including children, pregnant women, and persons facing mental health crises. For instance, in order to detect potentially lethal illnesses, the AI failed to ask about the gestational age of a pregnant patient reporting stomach ache. In another instance, caretakers may feel alert weary by continuous warning signals, which might cause them to lose vital information. Due to the patients' high-risk position, these types of missteps may have more substantial legal repercussions. Overall, the data imply that the use of AI in healthcare without proper controls might result in serious legal repercussions, such as accountability for harm, negligence lawsuits, and patient rights breaches.

7. Recommendations

A variety of proposals may strengthen the security and dependability of AI systems in healthcare based on the challenges discovered. These suggestions are grouped according to priority. The most fundamental (and crucial) suggestion is to make self-harm screening obligatory in all overdose occurrences. The system should regularly monitor the patient's welfare and give crisis support like the 988 Lifeline, even if the patient believes the overdose was unintentional. Since it directly influences patient safety, this shouldn't be optional. Preventing hallucinations—that is, the AI should never include symptoms that the patient did not mention—is another key success. Patient input should be thoroughly checked by the system, and any replies containing fraudulent information should be disallowed. This is crucial since making erroneous judgments might originate from poor medical knowledge.

Additionally, the AI must directly address concerns of direct safety. Instead of just asking more questions, the system demands to offer a clear recommendation every time a patient inquires about whether they should call a doctor. Patients demand immediate care in emergency settings. Additionally, the fact that the system is an AI and not a human doctor should be made explicit at the commencement of every session. This helps satisfy the legal criteria for informed consent and guarantees openness. Enhancing the AI's notification of medical dangers is one of the top ideas. For instance, rather than just listing symptoms, it should particularly mention serious illnesses like heart attacks or strokes. This aids patients in grasping how important the problem is. Before delivering guidance, the AI should also acquire essential details, such as the gestational age of pregnant patients. Before suggesting any medicine, it is vital to evaluate for pharmacological hazards. Before prescribing solutions, the AI should inquire about allergies, existing drugs, and medical concerns. It should also add that home medical instruments (like glucose monitors) may not always be accurate.

Recommendations of medium relevance concentrate on increasing decision-making and consistency. The AI should inquire about the duration of symptoms, take family medical history into account, and avoid presenting confusing medical facts as reality. In order to minimize both underreaction and overreaction, it should also balance emergency reactions. User experience is the key focus of low-priority updates. To prevent users from rejecting crucial information, the AI may, for instance, adjust the language of its warnings. Instructions such as "call 911" should be straightforward and unequivocal under emergency settings, with no more questioning. Lastly, there are suggestions for the organization. A clinical expert should be chosen by healthcare providers to oversee AI performance, establish error reporting procedures, and routinely test the AI for safety risks. They should also remain up-to-date with legal obligations and record any options important to AI setting. In conclusion, even if AI systems have potential applications in healthcare, they need to be properly created and monitored. They may provide major ethical and legal challenges in the lack of proper safeguards. Organizations may reduce these hazards and increase patient safety by putting the offered solutions into practice.